GNEG590V Assignment # 2

1. Please briefly describe the advantages of digital communication system over analog communication system.

2. A signal $s(t)$ can be written in a form as

$$s(t) = 3 + sin(2\pi t) + sin(100\pi t + 20) + cos(300t + 125). \quad (1)$$

   (a) What is the bandwidth (or the highest frequency component) of the signal?

   (b) If we transmit this signal in a channel with ideal bandwidth of 55 Hz (no attenuation from 0 to 55 Hz), will the signal be distored by the channel?

   (c) compare the properties of fiber optic cable an copper wire (list the advantages and disadvantages).

3. 2-3, 2-4, 2-5, 2-11